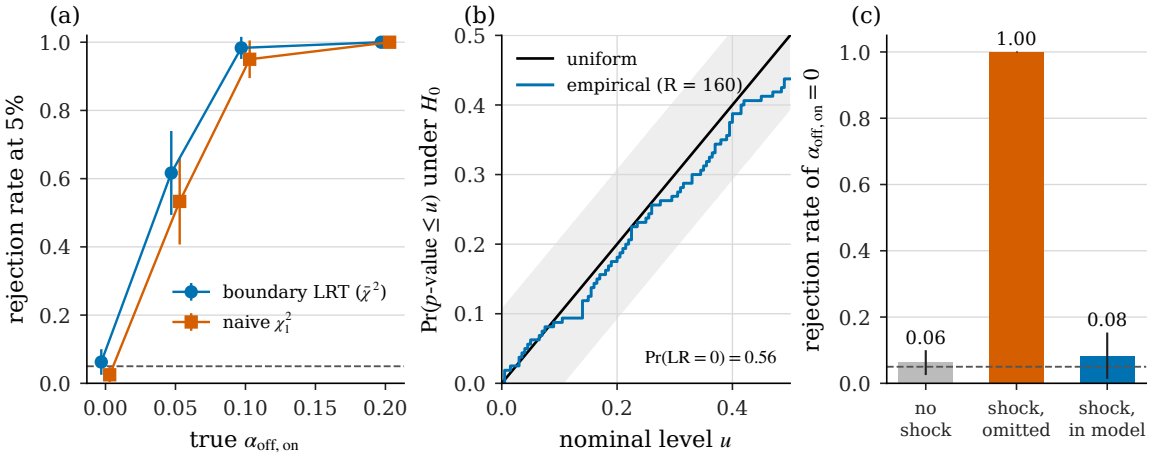
(a)
rejection rate at 5%
1.0
0.8
0.6
0.4
0.2
0.0
boundary LRT ($\bar{\chi}^2$)
naive χ^2_1
0.00
0.05
0.10
0.15
0.20
true $\alpha_{\text{off, on}}$
(b)
Pr(p-value $\leq u$) under H_0
uniform
empirical (R = 160)
Pr(LR = 0) = 0.56
0.0
0.1
0.2
0.3
0.4
0.5
nominal level u
(c)
rejection rate of $\alpha_{\text{off, on}} = 0$
1.00
0.06
0.08
no shock
shock, omitted
shock, in model